Optimum Design of Ship Structures

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Summary

Nonlinear programming techniques have recently been used rather extensively in conjunction with automated design and optimization of ship structures. This paper presents an introduction to the field and discusses the methods used, with special emphasis on the sequential unconstrained minimization technique (SUMT).

In practical design problems the number of free variables easily becomes high and the structural analyses may also become complex and time consuming. Large computer times are a consequence. Various methods by which the required number of redesigns during the search for optimum, as well as the computer work involved in each such redesign may be reduced are discussed. Approximate behaviour and constraint models in terms of simple polynomials have been used with considerable success and may be combined with an extrapolation technique that allows one to reduce significantly the number of response surfaces. The application of gradient methods of search combined with finite difference approximations of the gradients is demonstrated to yield efficient algorithms. Sequences of suboptimizations have also been tried with considerable success. In this manner the required computer time may by reduced to only a fraction of that previously experienced.

A few examples of typical applications to practical ship design problems are presented, viz.:

a) Optimum design of the midship section of an OBO-carrier.

b) Minimum weight design of grillages and transverse frames of tankers.

1 Introduction

During the first fifteen years of the computer-age great efforts have been made in order to computerize the analyses of engineering structures. It is, however, only recently that comparable efforts have been put into the more important and difficult field of computer aided design. Recent developments on time sharing systems combined with remote terminals using teletypewriters as well as graphical devices have stimulated tremendously the interest in man-machine modes of operation, by which the user may interact with the computer and thus more successfully use it as a design tool. Computer capabilities exist already today that allow the designer to solve structural problems by communicating with the computer through graphical displays |26|. It must be expected that this type of computer usage becomes increasingly important in the years ahead. It is, indeed, believed that most of the complex design problems of practical life can be solved efficiently only through some sort of direct interaction between computer and designer, at least in the foreseeable future.

In spite of this, we are presently also experiencing a rapidly growing interest in fully automated design capabilities that only require the designer to state his problem and thereafter expect the computer to do, more or less, the complete design. There is no conflict in these apparently opposing trends. The future designer will, while interacting with the computer, want to administer a library of programs, that automatically perform various parts of the design job. The task of the designer himself should preferably consist of supervision, establishment of priorities between conflicting requirements and other types of decision-making.

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This paper deals with some aspects concerning the development of fully automated design capabilities, and presents examples of practical applications in the field of ship structures design. It is understood that such automated design capabilities should be able to find the best structure, judged by a certain prescribed criterion. Hence any feasible solution is not accepted as the final design.

2 Formulation of the Design Problem

Before presenting the technique of automated design and optimization it is necessary to state the design problem in a suitable form. This will be accomplished with reference to Fig. 1, in which the most important steps of the design procedure are outlined |18|. At the outset the designer should always establish a clear understanding of the functions to be performed by the structure (I)[1]. This step includes an evaluation of the types and intensities of loadings, and other service requirements (such as: largest possible deck areas in the holds uninterrupted by pillars, bracketless frames).

Next the topology of the structure must be selected (II). The designer of a transverse bulkhead must decide whether the bulkhead should be plane or corrugated and also whether it should span vertically between horizontal stringers or horizontally between vertical webs.

When the general topology of the particular structure under study has been selected, it may be completely described by means of a certain number of design variables (III):

$$\boldsymbol{x}=(x_1, x_2, \cdots, x_n)$$

These quantities may represent dimensions of the structure (such as diameters, lengths, thickness, etc.) or any other physical property for which an independent choice may be made, such as yield stress of the material, etc. The behaviour of the structure under a certain loading condition may be characterized by means of t behavior variables (III):

$$\boldsymbol{\sigma}=(\sigma_1, \sigma_2, \cdots \sigma_t)$$

Such as stresses, deflections, elongations, etc. The behaviour variables are functions of the design variables as well as the loadings. For any well formulated problem there is always a unique vector $\boldsymbol{\sigma}$ corresponding to a given vector $\boldsymbol{x}$.

The following relationships between $\boldsymbol{x}$ and $\boldsymbol{\sigma}$ are representing the conventional analysis of the system (IV):

$$h_j(x_1, \cdots x_n, \sigma_1, \cdots \sigma_t)=0 \qquad j=1, 2, \cdots t \tag{1}$$

Similar relationships may be established for other loading conditions. Hence, to each selected set of design variables there exists a corresponding set of behavior variables for each loading condition. But some of these sets of variables $(x_1, \cdots x_n, \sigma_1, \cdots \sigma_t)$ cannot be accepted because they represent designs which are not permitted or undesirable. Such designs are excluded by means of different types of requierments which all may be written in the following form (V):

$$g_i(x_1, \cdots, x_n) \geq 0, \qquad i=1, 2, \cdots, m \tag{2}$$

These restrictions or constraints exclude unwanted solutions such as those corresponding to dimensions, stresses, deflections, etc., which are too small or too large, wrong geometrical proportions, instability of elements or of the structure as a whole, and so forth.

All the solutions which simultaneously satisfy relations (1) and (2) correspond to acceptable or feasible designs. Generally there will exist a large number of such feasible designs. These designs are, however, by no means equivalent. Judging by a certain criterion (V) it is likely that one solution will be better than any one of the others. This is than the optimum solution.

Before it is possible to start the search for the optimum it is necessary to formulate this criterion or object function, by which some of the feasible designs are preferred to others. This criterion may

1) Roman numerals in parentheses refer to the blocks of Fig. 1.

be expressed in terms of weight of the structure, cost of production or any other suitable quantity.

The criterion may in the present case be written as a functin $f(\boldsymbol{x})$ of the design variable $x_1, x_2, \cdots, x_n$.

When the designer has reached this stage he will be facing a computational task which may be both time consuming and tedious, consisting of the following steps (VII~XII):

1. Select design variables.
2. Analyze the system by solving equations (1) in order to find corresponding behavior variables
$$\sigma_k = c_k(x_1, \cdots, x_n) \qquad k=1, 2, \cdots, t$$
3. Control if the solution is feasible, if so go to 4, if not go to 5.
4. Evaluate the criterion $f(\boldsymbol{x})$.
5. Change design variables and repeat steps 2 to 5 with the ultimate goal of finding the design that corresponds to $f(\boldsymbol{x})=$ minimum (or sometimes $f(\boldsymbol{x})=$maximum).

Today it is possible to do this work by means of the electronic computer, as will be discussed in the following sections. Having found the optimum solution the designer must, of course, himself undertake the final judgement of the solution obtained, (XII), and decide whether it is necessary or desirable to go back to step (II) in the design process and change the topology in a search for better solutions. Through the optimization of several different topologies the chances of finding the true optimum are obviously increased.

Some of the available methods of mathematical serch for the optimum solution of a problem of the type stated above will be discussed in the following sections. Before entering into that problem, it may, however, be appropriate to present in detail an example which includes a very simple optimum design problem, formulated according to the outline given above.

Example 2.1 Minimum Weight Design of a Hatch Cover |18|

A hatch opening of width $l_0=6.0$ m, see Fig. 2a, is to be covered by 600 mm wide box beams of aluminum with cross-sections as shown in Fig. 2b. The hatch cover is subjected to a maximum loading of 1000 kp/m^2. The flange thickness (t_F) and the height (h) should be selected such that the weight of the box beams is a minimum.

The following requirements should be met:

1) and 2) h and t_F should be positive
3) Maximum shearing stress $\tau \leq \tau_{all}$
4) Maximum bending stress $\sigma \leq \sigma_{b,all}$
5) Buckling of the flanges must not occur, $\sigma_b < \sigma_k$
6) Maximum deflection $\delta < K \cdot l_0$

These requirements may, in accordance with equation (2), be written as follows:

$$g_1 = h > 0$$
$$g_2 = t_F > 0$$
$$g_3 = \frac{\tau_{all}}{\tau} - 1 \geq 0$$
$$g_4 = \frac{\sigma_{b,all}}{\sigma_b} - 1 \geq 0$$
$$g_5 = \frac{\sigma_k}{\sigma_b} - 1 \geq 0$$
$$g_6 = \frac{Kl_0}{\delta} - 1 \geq 0$$

Expressions are next introduced for τ, σ_b, σ_k and δ in terms of t_F and h, yielding all restrictions as functions of the free variables only |18|.

All of the restrictions listed above are plotted in Fig. 3 for the set of numerical values listed in

the figure. Desingns corresponding to points situated above and to the right of the hatched portions of curves g_5 and g_6 are fearsible.

For the purpose of simplicity, the weight of transverse bulkheads and stiffeners is neglected, and the object function is written as follows:

$$f(t_F, h) = W/\gamma l_0 = 2t_s h + 2t_F b = h + 120 t_F$$

where W is the weight of the box beam.

If the weight function f is plotted along the axis perpendicular to the $t_F - h$ plane, a slanting plane passing through the origin is obtained. Some lines of constant weight are shown in Fig. 3. In the present example the optimum solution is found in the point of tangency between the restriction g_5 and the weight line corresponding to $f=101$. Hence, the restriction against buckling is active, but the ptimum solution is also quite close to the restriction on deflection (g_6).

3. Problem Identification

Mathematically the design problem just stated may be formulated as follows :

$$\begin{aligned} &\text{Find } \bar{x} \text{ such that} \\ &\quad f(x) \text{ is minimized} \\ &\text{subject to } g_i(x) \geq 0 \qquad i = 1, \cdots, m \end{aligned} \tag{3}$$

The vector $\bar{x}$ then represents the optimum feasible design.

In order to visualize characteristic features of this type of problems, graphical representations of some additional very simple design problems, with only two free variables ($n=2$), are presented in Fig. 4. In all of these examples weight is assumed to the criterion.

The n-dimensional spanned by the free variables is called the design space. A given combination of the free variables x is called a design point.

The constraints can be visualized as hyper-surfaces in the n-dimensional space (hyperplanes for linear constraints). In the two-dimensional examples of Figs. 3 the 4, the constraints appear as curves and straight lines. A constraint (i) is called active in a design point for which $g_i(x)=0$ and passive if $g_i(x)<0$. If $g_i(x)<0$ the constraint is violated and the corresponding design is unacceptable.

Formulations in which the object function $f(x)$ or any of the constraints $g_i(x)$ are nonlinear, yield nonlinear programming problems. Even the simple design problems presented in Figs. 3 and 4 are nonlinear. The constraints are nonlinear while for three of the examples object functions in terms of weight are linear.

A commonly used technique to solve simple design problems is to assume that a selected set of constraints are active at the optimum point. These constraints are then treated as equalities and are used to eliminate free variables. The number of active constraints may generally be at the most equal o the number of free variables. If the number of active constraints just equals the number of free tvariables, one generally has a socalled "fully stressed design". Numerous examples of the approach just mentioned may be found in the literature on minimum weight design. If the constraints are nonlinear this approach may become impracticable for all but the simplest cases. The design found by this approach will be located at the intersection between the set of selected active constraint surfaces. While this approach may yield the optimum solution in some cases, such as the one shown in Fig. 4 B, Fig. 4 A demonstrates a different situation. Here the weight of the fully stressed solution (W_1) exceeds that of the optimum solution (W_0) by more than fifty percent. A situation such as the one shown in Fig. 3, where the fully stressed solution is almost as good as the optimum solution may be more typical for many structural problems. Techniques which seek fully stressed solutions should, however, be used with extreme care.

In a general design problem there is no simple way of knowing which constraints are active at the

optimum point. The total number of constraints is in more structural problems much higher than the number of free variables.

An optimum solution $\bar{x}$ is characterized by the property that there are no other feasible designs x in the immediate neighbourhood of $\bar{x}$ that correspond to a lower value of the object function than $f(\bar{x})$. Mathematically this can be expressed by the Kuhn-Tucker condition :

$$\frac{\partial f(x)}{\partial x_i} - \sum_{j=1}^{m} u_j \frac{\partial g_j}{\partial x_i} = 0 \qquad i=1, \cdots, n \tag{4}$$

$$u_j g_j(x) = 0 \qquad j=1, \cdots, m \tag{5}$$

$$u_j \geq 0 \qquad j=1, \cdots, m \tag{6}$$

In Fig. 4 B a geometric interpretation of the Kuhn-Tucker condition is presented. It simply states that the gradient to the object function $Vf(x)$ must pass through the cone generated by the gradients to the active constraints (here $Vg_1(x)$ and $Vg_2(x)$). If this condition is not fulfilled there will always be a feasible direction along which it is possible to improve the abject function $f(x)$.

It is important to note that the Kuhn-Tucker condition is valid only for convex nonlinear programming problems. Figs. 4 A-B are convex problems since straight lines joining any two feasible points lie entirely in the feasible subspace. This is not, however, true for the problem depicted in Fig. 4 C, which is therefore termed a non-convex programming problem. In this case there are three solutions that all satisfy the Kuhn-Tucker condition. They all correspond to relative or local minima. The best solution, W_0, is called the global optimum. While most of the nonliner programming algorithms will converge for convex programming problems, one can for non-convex problems at best expect convergence to a relative minimum. Also a fully stressed iteration procedure may fail in such cases since, even if the optimum design is fully stressed, this may also be true for several of the other designs corresponding to local minima. This is actually the case for the example in Fig. 4 C.

4. Optimization Methods

Several rather different techniques have been applied to solve structural design problems of the type stated by Eqs. (3). Moses |20|, Cornell, Reinschmidt and Brotchie |6|, and Romstad and Wang |25| have developed techniques based on the application of the well known Simplex method of linear programming. They approximate the nonlinear programming problem by a sequence of linear programming problems. Gellatly and Gallagher |10|, Brown and Ang |4|, and Moses and Onoda |21| have all used methods of "feasible directions" to solve structural design problems. In these methods the search is first directed towards the boundary of the feasible region and later continued roughly speaking along the boundary.

A third category of nonlinear programming methods is based on the socalled penalty function techniques. Schmit and Fox |27| used an exterior penalty technique, while Kavlie, Kowalik and Moe |12| and Marcal and Gellatly |17| applied an interior penalty technique.

Moses and Onoda |21| have demonstrated that the elastic design of grillages generally leads to non-convex programming problems. Kavlie and Moe |13| have found penalty function techniques to have some very desirable features for such problems. These approaches are as a whole of great interest for general automated design routines. In the following only penalty function techniques are dealt with, but, of course, this does not mean that other techniques are not applicable in structural design. In a recent paper Abadie |1| claims that his generalized reduced gradient (GRG) method is more efficient than the penalty technique described in the following. Abadie's GRG method is of the feasible direction type. His comparison is based on a few sample problems.

The basic idea of the penalty function techniques is extremely simple. The constrained problem stated by Eqs. (3) is transformed into an unconstrained one by the addition to the object function of a

penalty term which takes care of the effect of the constraints. An unconstrained minimization may then be performed on the transformed function, which generally is written as follows :

$$P(x, r_k)=f(x)+r_k\sum_{i=1}^{p} G(g_i(x)) \tag{7}$$

The surface described by the P-function is often termed response-surface and the r_k-value is called the response factor. The second term of Eq. (7) is the penalty term.

If the minimization is repeated for a sequence of values of the response factor, the solution may be brought to converge to that of the constrained problem. The minimization of an unconstrained function such as $P(x, r_k)$ is considerably simpler than to solve the original problem. Unconstrained minimization is discussed in section 5.

Some different choices for the penalty term $G(g_i(x))$ are demonstrated in Fig. 5 for the simple problem :

$$\min ax, \text{ subject to } g_1(x)=x-b\geq 0 \tag{8}$$

A penalty function approach similar to the one used in this paper hwas first suggested by the 1969 winner of the Nobel prize in Economics, Frisch |9|. His "logarithmic potential method" employs the function :

$$P(x)=f(x)+\sum_{i=1}^{m}\alpha_i \ln g_i(x) \tag{9}$$

Frisch did not, however, perform a sequence of unconstrained minimizations.

Ablow and Brigham |2| suggested transformations of the types shown in Figs. 5 B and C. In 1961 Carroll |5| proposed the form of the penalty function which has been used in this paper and seems to be the most popular one (Fig. 5 A):

$$P(x, r_k)=f(x)+r_k\sum_{i=1}^{m} 1/g_1(x) \tag{10}$$

This P-function is repeatedly minimized for a sequence of decreasing values of r_k.

Fiacco and McCormick |7| have contributed considerably to the development of this approach, which they call "sequential unconstrained minimization technique" or SUMT.

It is aparent from Fig. 5 A that Eq. (10) yields an interior penalty function while the penalty function shown in Fig. 5 C is exterior. Eq. (10) requires a feasible starting oint for the search toward optimum. It does not allow any constraints to be violated. The penalty term is simply not defined if x is non-feasible. For some design problems it may be rather difficult to find a feasible initial design. When approximate methods of analysis are used some intermediate design points during the search for optimum may also be found to be non-feasible. One therefore would like to have a procedure that in such cases draws the design back to the feasible region. In Fig. 5 D an extended penalty function is presented that satisfies this requirement |22|. It is actually a combination of the penalty functions of Figs. 5 A and B is defined as

$$G(g_i(x))=\begin{cases}1/g_i(x) & \text{for } g_i(x)\geq\varepsilon\\ (2\varepsilon-g_i(x))/\varepsilon & \text{for } g_i(x)<\varepsilon\end{cases} \tag{11}$$

were $\varepsilon=r_k/\delta$

δ=constant that defines the transition between the two types of penalty terms, see Fig. 5 D. One may for instance tentatively select δ approximately equal to $f(x)$.

This extended penalty function does not complicate the computational procedure. Fig. 6 illustrates a typical appearance of three consecutive P-functions (Eq. 10)) for a grillage design problem as shown in Fig. 7. In this problem there are three distinct relative minima |22|.

5 Unconstrained Minimization

The selection of an efficient method for unconstrained minimization becomes extremely important in

SUMT, since a sequence of such minimizations has to be performed for gradually decreasing response factors. There is a wide choice of methods of search. In directed methods the minimization is usually performed by successive unidirectional searches in a number of different directions in the following manner :

$$x^{i+1}=x^i+\lambda_i s^i \tag{12}$$

were $s^i=i$-th direction vector

$\lambda_i=i$-th step length

$x^i=$starting value for the i-th step

$x^{i+1}=$the design vector corresponding to the minimum of $P(x)$ along the current direction s^i

$P(x)=$the unconstrained function to be minimized.

Among the methods of directed search one may, according to the manner in which the directions s^i are determined, distinguish between methods of "direct search" which are making use of only function values $P(x)$, and "gradient methods" which also require the computation of gradients of $P(x)$:

$$\nabla P(x)=\left\{\frac{\partial P(x)}{\partial x_1}\cdots\cdots\frac{\partial P(x)}{\partial x_n}\right\} \tag{13}$$

Kowalik |15| has described in detail a direct search method due to Powell |23| as well as a method using conjugate gradient directions due to Fletcher and Powell |8|.

The principle of Powell's direct search method is demonstrated in Fig. 8. Having in this example only two free variables, x_1 and x_2, it is possible to visualize the search for the best solution. From an arbitrarily chosen starting point (x^{01}) the search at first starts out along a straight line, which in the present case is taken paralled with the x_1-axis. The minimum (x^{11}) along this line is determined. From x^{11} the search next proceeds along a line parallel to the x_2-axis, until a new minimum (x^{21}) is found. The third search is carried out along the vector ($x^{21}-x^{01}$) and leads to a new minimum denoted x^{02}. Applying the two last used directions and x^{02} as a new starting point, the process described above is now repeated, yielding at the end of the second cycle x^{03} which is again the starting point for the third cycle.

Returning back to the hatch cover example of section 2 Fig. 9 demonstrates how this problem is solved by means of the SUMT technique in connection with Powell's method of direct search. The minima $M_1, M_2, \cdots, M_5$ corresponding to consecutive response surfaces are shown and also the search pattern which is followed. Note that one of the two starting points is unfeasible. A special algorithm has been devised which first checks whether the starting point is feasible. If this is not true, the g-functions (Eq. 2) which are negative are maximized until they become positive, thereby yielding a feasible starting point. It is very convenient in practical optimization studies thus to be able to start with an arbitrarily chosen point without having previously to ensure that it corresponds to a feasible solution.

One important task is to determine that value of λ (Eq. (12)) which yields minimum of the P-function in the current direction of search. A very simple but rather inefficient procedure that works for any unimodal function is the Golden Section method |15|. By this method an interval containing the minimum is first determined. Next this interval is gradually reduced in a systematic way.

In the function to be minimized has a smooth variation, approximations by simple polynomials have been suggested by Powell |23| who uses quadratic polynomials in connection with direct search, and by Fletcher and Powell |8| who describe the use of cubic polynomials in combination with their gradient method.

All of these approaches have certain drawbacks when used for minimixation of the P-function. The Golden Section method requires as many as 8-10 function evalutions along each direction. Even when

efficient reanalysis techniques are used this is not very attractive for statically indeterminate structures requiring a substantial numerical effort at each design point.

A direct polynomial approximation to P does not work very well since the function considered is not easily approximated by simple polynomials (Fig. 5) in the neighbourhood of the constraints.

Lund |16| has described an efficient scheme used in combination with the direct search method of minimization (see also |19|). Instead of approximating P directly by simple polynomials, each of its components, that is the object function and all of the constraints, are individually approximated. These functions generally vary quite smoothly with no singularity at $g_i(x)=0$. This approach has been found to work well also in connection with the gradient method.

In a recent paper Powell |24| has presented a new gradient method based on the theory of quadratic functions in which on such unidirectional searches are required. The direct search method of Powell as well as the conjugate gradient method of Fletcher-Powell have been used in connection with SUMT to a wide range of problems in structural optimization. Based on this experience as well as a study by Box |3| it appeares that if the computational effort to determine the gradient corresponds atmost to n function evaluations, Fletcher-Powell's gradient method is generally slightly superior to Powell's method of direct search. These two methods should represent good choices within their groups.

One other important practical consideration is the additional effort required if analytical expressions for the gradients need to be programmed. Gradient directions may also be derived by means of finite difference schemes |29, 22|. This was done in the program used for the two last numerical examples described in scction 7.

6. Application in Ship Structures Design with Rule Constraints

Nonlinear programming techniques of the types just described have been applied to several practical design problems in the shipbuilding industry. The goal of most of these studies has been to find structural configurations while meeting the strength requirements established by the pertinent Classification Society, yield the lowest possible cost of production. Considerable efforts have been made in order to obtain reliable and detailed information about contributions of various items to the cost of production. Close cooperation with shipyards has been maintained during this part of the study.

It is a characteristic feature of many such design problems that the rule requirements are explicitly stated in terms of formulae for minimum cross sectional properties of the structural elements. Hence, the structural analysis equations (1) may be non-existent or, indeed, very simple. The time required for each function evaluation is therefore rather moderate. Structural optimization is of particular significance in connection with the construction of super-tankers and large bulk-carriers. Fifty percent or more of the total cost of production may for some these ships be due to the hull structure alone. As initial costs may account for a very substantial part of the total expences of ship operation, cost of hull construction may be by far the biggest single item to attack in an attempt to reduce the shipowners cost of operation. A major part of the steel weight built into such ships is spent at the moment when the structural configuration of the midship section is decided upon. Programs have been developed that automatically seek to optimize the midship section of such ships, and sysmatic analyses of effects of variations in the relationships between cost of workmanship and materials, principal dimensions of the ships etc. have been reported |18|. An example of such a study in presented in the following Example 6.1. For the very heavy structures in the biggest ships it has been found that cost of materials is the most important item. In such cases the accuracy in cost estimation is not a too critical factor, and the optimum solution tends towards a minimum weight design. In |11| the design of a car-carrier with many relatively light decks is discussed. Here cost of labour is an important item, and the design corresponing to minimum cost of production is radically different from that

coresponding to minimum weight.

Example 61-Optimum Design of Midship Section of OBO-Carrier.

Powell's method of direct search has been applied to study optimum midship scantlings of OBO-carriers with topology as indicated in Fig. 10.

Presentation of the design problem

The program is based upon the following assumptions :

1. Principal dimensions of ship are given as input.
2. Positions of transverse bulkheads, shapes of wing tanks, sizes of hatch openings as well as maximum still water bending moments are given as input.
3. The hold considered carriers full load with adjacent holds empty.
4. Scantlings have to comply with the 1967 rule requirements of Det norske Veritas.
5. Double bottom is to be designed by means of the theory of orthotropic plates.
6. Plate thicknesses and frame spacings may vary continuously.
7. For the frames used, cross sectional areas (A) may be expressed as continuous function of the section moduli (Z). The following relationships are assumed :

   Slabs in deck : $A=1.18\,Z^{0.64}$

   T-sections in main transverse frames : $A=0.928\,Z^{0.601}$

   Bulb-sections in top wing tanks and doulbe bottom : $A=1.57\,Z^{0.53}$

The design problem involves the following eight free variables :

x_1=Thickness of plates in inner bottom

x_2=Thickness of plates in outer bottom

x_3=Spacing between longitudinal girders in double bottom

x_4=Spacing between transverse girders in double bottom

x_5=Height of double bottom

x_6=Thickness of plates in deck

x_7=Spacing between longitudinal frames in deck

x_8=Section moduli of frames in deck

The spacing between transverse main frames is assumed equal to half the distance between transverse girders in the double bottom. Plate thickness and section moduli of frames in ship side and bottom of top wing tank are explicit functions of the frame spacings and need not be considered as free variables. The same applies to frame dimensions in the double bottom.

Constraints

Altogether twenty constraints are involved, that account for

a) local strength of plates in the double bottom,

b) local strength of deck longitudinals,

c) local strength of the double bottom regarded as an orthotropic plate,

d) longitudinal strength of the ship hull girder,

e) other design requirements such as minimum or maximum dimensions etc.

Object function

Weight as well as cost functions have been included in the program. The weight function includes practically all of the scantlings except for the transverse bulkheads.

Results

A number of systematc studies have been performed. As an example the results of three runs related

to weight minimization of a 128000 t.dw. OBO-carrier will be presented here. The principal particulars about the ship considered are as follows :

Length	$L=280$ m
Breadth	$B=41.1$ m
Draught	$d=15.55$ m
Length of hold	$l=19.6$ m
Breadth of hatch opening	$b=17.0$ m

The following alternatives are considered :

A) Depth moulded $D=20.8$ m, height of double bottom free

B) Depth moulded $D=20.8$ m, height of double bottom 2.1 m

C) Depth moulded $D=21.2$ m, height of double bottom 2.5 m

Note that the hold capacities are equal for cases B and C.

The results of the three runs are presented in Table 1. In these runs mild steel has been assumed throughout. High strength steels may also be considered.

Table 1 shows that large plate thicknesses are required in the deck in order to satisfy the longitudinal strength requirements. This is due to the relatively large still water bending moment considered, the large L/D ratio (13.2—13.5) and the wide hatch opinings. From the point in weight it is advantageous to put the longitudinal strength material as distant as possible from the neutral axis. Hence, the relatively thin plates in the bottom of the top side wing tank. These plates satisfy the local strength requirements.

Alternative C shows that 27 tons of steel may be saved per hold section, in comparison with alternative B, by increasing the depth moulded by 0.4 m. This saving amounts to approximately 300 tons for the entire ship, corresponding to approximately 2 percent of the total hull weight. Similar results have been obtained through cost minimization studies. For each alternative less than one minute of computer time was required on a Univac 1107 computer.

7. Optimum Design of Statically Indeterminate Structures

7.1 Introduction.

The example just presented was particularily simple since all the rule requirements could be formuated explicitly in terms of the design variables or as simple functions of these. Many design problems obviously require a more complex treatment. Rational design procedures based on comprehensive structural analyses, usually involving complex statically indeterminate systems, and specified stress levels, are today rapidly replacing the more empirical approaches.

Generally speaking the aim is to develope automated procedures by which to select the cross sectional properties of the members of systems of beams and frames. Some typical cross sectional forms are shown in Fig. 11. Each cross section may be described by a number of free variables, typically one to four.

In practice one does not always want all of the frame members to be different. Hence, it may be advantageous if several members can be described to have identical cross sectional shapes. This can be accomplished by means of a member type array which links each member cross section to one of the available cross sectional shapes. The frame of Fig. 12 may for instance have a member type array such as

$$\boldsymbol{J}=\{1,1,3,3,2,3,3,2,4,4,5,5\} \tag{14}$$

which simply expresses that :

members 1 and 2 are of type 1, 5 and 8 of type 2, 3, 4, 6 and 7 are of type 3, etc.

With three variables for each member type (as in Figs. 11 A and B) one would for this frame have altogether $n=3\times5=515$ free variables.

7.2 Method of Analysis.

The displacement method of structural analysis has been used to determine the behaviour of a given design under prescribed loading conditions. The program developed can analyze any type of three-dimensional frame structure for which the members are parallel to the three coordinate planes. Lateral bending and torsion of the members are neglected. Axial deformations can be included.

One of the major tasks in the analysis is to solve the equations of equilibrium :

$$\boldsymbol{K}\boldsymbol{r}=\boldsymbol{R} \tag{15}$$

where $\boldsymbol{K}$ = the system stiffness matrix
$\boldsymbol{r}$ = the vector of nodal point displacements
$\boldsymbol{R}$ = the vector of nodal point loads.

Efforts have been made to develop a quite efficient analysis program, taking into consideration the variable band width of the stiffness matrix |14|.

For structures of practical importance the total number of members and hence the number of stress constraints will usually be quite large. For the longitudinal frame of a tanker discussed in Example 7.3.3 of the next section, with 31 members, 3 loading conditions and with upper and lower bounds on three normal stresses and two shearing stresses for each member, there are altogether 930 stress constraints. There are on the other hand only 27 free variables. At the optimum solution there are in most cases fewer active constraints than free variables. Only the active constraints are of real importance to the design algorithm. It would be a waste of numerical effort and storage if all the stress constraints were included in the penalty term. A simple and practical way to avoid this is to specify in the input which stresses should be included, on the basis of computed constraint values at the starting point. At the end of each unconstrained minimization it should then be checked whether any of the constraints not included are violated.

A more sophisticated program that also has been used, lets the computer select the active constraints on the basis of a check of all constraint values at specified intervals. The only difficulty in this approach is that for the intial response surface most of the constraints may be of the same order of magnitude. Hence it may be difficult to decide at this stage which should be included in the penalty term.

Several of the discussed search procedures involved analyses of a number of slightly modified structures obtained by the addition or subtraction of Δx_i subsequently for each free variable (x_i). Such analyses can be performed with much less computational effort than that required in complete analyses. For the modified structure the following equation of equilibrium is valid :

$$(\boldsymbol{K}+\Delta\boldsymbol{K})\boldsymbol{r}_m=\boldsymbol{R} \tag{16}$$

$$\boldsymbol{r}_m=\boldsymbol{r}+\Delta\boldsymbol{r} \tag{17}$$

where $\Delta\boldsymbol{K}$ = the change in the structure stiffness matrix corresponding to the modification of the i-th variable Δx_i
$\Delta\boldsymbol{r}$ = the corresponding change in the displacement vector
$\boldsymbol{r}_m$ = the modified displacement vector.

Substitution from Eq. (17) into Eq. (16) and rearranging yields

$$\boldsymbol{K}\boldsymbol{r}_m=\boldsymbol{R}-\Delta\boldsymbol{K}\boldsymbol{r}-\Delta\boldsymbol{K}.\Delta\boldsymbol{r} \tag{18}$$

A linearized approximation $\boldsymbol{r}_{mi}$ to $\boldsymbol{r}_m$ is obtained by neglecting the last term of Eq. (18):

$$\boldsymbol{K}\cdot\boldsymbol{r}_m=\boldsymbol{R}-\Delta\boldsymbol{K}\boldsymbol{r} \tag{19}$$

or by means of Eq. (15):

$$\boldsymbol{K}\Delta\boldsymbol{r}_i=-\Delta\boldsymbol{K}\boldsymbol{r} \tag{20}$$

The stiffness matrix $\boldsymbol{K}$ has already been decomposed when determining $\boldsymbol{r}$. Hence each perturbation only requires the reduction and backsubstitution of a new set of modified right hand sides of the type

shown in Eq. (20).

Only the members which are defined by the free variable x_i will contribute to ΔK. The vector J (Eq. (14)) is used to find these members.

Substantial additional savings in the computational work have been achieved by means of a technique employing approximate behaviour models |19|. This technique takes advantage of the fact that in statically indeterminate structures the changes in generalized member forces caused by variations in member sizes are usually relatively small. Hence, it is possible to create simple and fairly accurate polynomial relationships between generalized forces and member sizes.

7.3 Numerical Examples

The numerical examples presented in this section are obtained employing Fletcher-Powell's gradient method of unconstrained minimization in combination with SUMT. The examples are originally presented in references |13| and |14|. A finite difference scheme is used to obtain the gradients in the last two examples. It should be emphasized that the numerical examples are only designed to demonstrate the capabilities of the method. A few adjustments would have to be made in order to make the program useful for practical purposes. These would include the use of effective lengths of members instead of distances between nodal points in calculating stiffnesses, calculation of stresses at critical sections rather than at nodal points, etc. Such modifications would not influence noticeably the efficiency of the automated search procedure.

Example 7.3.1-Grillage.

As a first example a three by three grillage problem with eight free variables studied by Moses and Onoda |21| will be considered, see Fig. 13. An interesting feature of this example is that Moses and Onoda found four distinctively different relative minima using different starting points. They used a method of feasible directions and also a fully stressed iterative design method. Only one variable, the cross sectional area, was used to define each member type. Moments of inertia and section moduli were derived from the areas using approximate empirical relationships. All of the members were subjected to a uniformly distributed transverse load of 1 kip/in. Upper and lower bounds on normal stresses of 20 kips/in^2 were prescribed. Using the method described in this paper the automated search routine was started from six entirely different points, see Table 2. In all of these cases did the solution converge to the best relative minimum, the global optimum. For this example the analysis of one design was executed in 0.023 seconds on the CDC 6400 computer. Gradient directions were calculated from analytical expressions in 0.052 seconds. When the Golden Section method of unidirectional search was used, the optimum was found in 12.9 seconds, with 472 function evaluations, and 48 gradient evaluations. Using polynomial approximations of the type described by Lund |16| in the unidirectional searches and a slightly changed convergence criterion, the same problem was solved in 5.4 seconds with 83 function and 35 gradient evaluations. In Table 3 the relative minima found by Moses and Onoda are presented together with the optimum found by the automated search routine described in this paper. Using SUMT the same minimum of the first response surface was found for all of the selected starting points. From this minimum on the search pattern was, of course, identical for all of the starting points.

Example 7.3.2-Transverse Tanker Frame |14|

As a next example a web frame of a 100,000 tons deadweight tanker is dealt with, see Fig. 12. This web frame was subjected to the four different loading conditions shown in Fig. 14. The twelve members are linked to the available five member types by the vector J presented in Eq. (14). While member of type 2 have symmetrical I-sections (Fig. 11 B), all of the other members have cross sectional shapes as shown in Fig. 11 A. Plate flange are given as input.

The allowable stresses are selected rather high to compensate for the fact that stresses are computed

at nodal points rather than at critical cross section :

$$\sigma_u = -\sigma_L = 1800 \text{ kp/cm}^2$$
$$\tau_u = -\tau_L = 1200 \text{ kp/cm}^2$$

In addition to stress constraints, limitations on plate thicknesses and flange areas are included.

Weight is selected as the criterion. A term that qualitatively accounts for the weight of web stiffeners is added whenever $H/t_W > 50$:

$$W_W = \begin{cases} \gamma A_W & \text{for } H/t_W \leq 50 \\ \gamma A_W [1 + 0,5 \cdot 10^{-4} (H/t_W - 50)^2] & \text{for } H/t_W > 50 \end{cases} \tag{21}$$

where W_W = weight of web per unit length
γ = specific weight of steel
t_W = web thickness

Table 4 presents the input dimensions as well as final values found by the automated search routine. Fig. 15 shows the resulting design and also the governing stresses. The experienced designer will notice immediately that buckling constraints have not been included here.

The complete optimization using finite difference derivatives and a model of analysis including both shear, axial and bending deformation of the members was performed in 75.2 seconds with 240 function and 62 gradient evaluations. Note that the initial point was not feasible, but only one unidirectional search was required to bring the design into the feasible region using the extended penalty function of Eq. (11).

Example 7.3.3-Longitudinal Frame with Intersecting Girders

Finally consider an additional problem from the 100,000 tons deadweight tanker of the previous example. The longitudinal frame with intersecting transverse girders in bottom, on a transverse bulkhead and under the deck, is shown in Fig. 16. Member numbers are indicated on the figure. The structure is symmetric with respect to the plane of the longitudinal frame. Nine different member type each characterized by three free variables, yield alltogether 27 variables. The member type array is as follows :

$$\boldsymbol{J} = \{2, 3, 3, 3, 3, 2, 1, 1, 1, 1, 1, 1, 4, 4, 4, 5, 6, 5, 6, 8, 9, 9, 9, 9, 8, 7, 7, 7, 7, 7, 7\}$$

Three loading conditions were considered. Transverse beams were assumed to be elastically supported at their ends. This three-dimensional frame was analyzed in 0.14 seconds. A gradient was found in 2.64 seconds. The complete problem was solved in 320 seconds with 361 function and 95 gradient evaluations. Table 5 presents the initial design and the final solution.

8. Suboptimization

The preceeding examples demonstrate that the method outlined can handle efficiently rather complex and realistic design problems. It is, however, obvious that required computer time may for other problems by far exceed those of the examples presented.

One way of reducing the computational effort is to reduce the dimensionality of the problem. One may for instance in some cases formulate several subproblems and subsequently take care of the coupling between the subproblems by means of a major optimization in which each subproblem is represented with fewer free variables.

If each member type of a frame is described by means of three free variables related to the cross sectional shape, it is natural to consider the selection of the optimum shape of each cross section as a subproblem leaving only one free variable describing the member size to be a free variable in the final optimization of the complete frame. Numerical examples obtained by means of such approaches are presented by Kavlie and Moe [22].

9. Conclusions

Some of the major conclusions obtained in the studies summarized here are :

1. The SUMT technique of nonlinear programming is well suited for automated design and optimization of ship structures.

2. Fully stressed iterative design procedures should be used with care since they may lead to designs which are far from optimum.

3. In the SUMT approach a proper choice of initial response factor may be very important when several local optima exist |22|.

4. The extended penalty function technique presented permits infeasible initial designs.

5. A special type of polynomial approximations applied in the unidirectional searches yield considerable savings in computer time.

6. Additional savings were obtained by means of efficient reanalysis techniques and a method employing "behaviour models".

7. The computational efforts have been reduced drastically for some numerical examples by means of judicious schemes of suboptimazation. It is suggested that these approaches should be explored further, bearing in mind that they may occationally lead to local optima.

8. The state of the art of this field is now such that automated design procedures can be used on quite realistic design problems. Practical implementation remains to be done.

Appendix I-References

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Appendix II-Notations

A = area of cross section
$f(x)$ = object function
$g_i(x)$ = i-th constraint on behaviour
H = height of cross section
I = moment of inertia of cross section
J = member type array
k = number of current response surface
$\boldsymbol{K}$ = system stiffness matrix
m = number of inequality constraints
n = number of free variables
$P(x, r_k)$ = transformed function
$\boldsymbol{r}$ = nodal point displacement vector
r = response factor
$\boldsymbol{R}$ = loading vector
$\boldsymbol{s}$ = vector of search direction
$\boldsymbol{x}$ = vector of free variable
Z = section modulus of cross section
γ = specific weight of material
λ_i = step length in directions s^i
σ = normal stress
τ = shearing stress

Subscripts :
f = flange
p = plate
w = web

Table 1 Results from Weight Minimization of O BO-Carrier

Alternative		A	B	C
Weight	(tons)	1300.5	1324.1	1296.9
Plate thickness, inner bottom	(cm)	2.05	2.42	1.83
Plate thickness, outer bottom	(cm)	2.73	3.06	2.74
Spacing of long. in d.b.	(cm)	191	181	196
Spacing of transv. in d.b.	(cm)	229	210	225
Height of d.b.	(cm)	232	210	250
Plate thickness, deck	(cm)	5.12	5.36	5.12
Spacing of long. in deck	(cm)	49.5	56.3	47.5
Sect. mod. of longitudinals	(cm^3)	2368	2449	2011
Plate thickness top wing tank	(cm)	0.99	1.13	0.95
Plate thickness, ship side	(cm)	2.62	2.62	2.57
Distance betw. transv. frames	(cm)	114	105	112
Frame spacing, inner bottom	(cm)	47.6	45.2	48.9
Framə spacing, outer bottom	(cm)	95.3	90.4	97.9

Table 2 Alternative Starting Points Used in SUMT (3×3 grillage)

Alternative	Values of free variables (x_1 to x_8)							
P	40	40	70	70	30	30	30	30
Q	30	50	45	70	10	12	10	12
R	40	40	40	40	40	40	40	40
S	60	60	15	15	40	12	12	12
T	15	15	60	60	12	40	40	40
U	5.5	5.5	5.5	5.5	50	60	50	60

Table 3 Final Solutions Obtained by Different Methods (3×3 grillage)

Design	Method	Values of free variables (x_1 to x_8)								Volume
A	I.R.	20.9	34.9	29.5	35.0	6.9	6.8	7.2	7.1	24.410
B	I.R. F.D.	5.0	5.0	5.0	5.0	29.9	34.9	34.9	41.0	27.660
D	I.R.	5.0	5.0	47.8	56.0	12.3	17.8	13.7	16.5	23.230
E	F.D.	5.1	5.1	47.5	55·8	13.0	17.0	13.6	16.6	23.190
F	F.D.	27.5	31.5	32.8	39.3	7.4	8.0	9.8	10.3	25.000
C	SUMT	5.1	5.1	47.2	55.3	13.5	16.6	14.7	15.9	23.184

(I.R.=iterative redesign, F.D.=feasible directions, SUMT=automated search routine described in this paper)

Table 4 Results of Tanker Web Frame Optimization (All dimensions in cm and cm²)

Member type	Plate flange area A_p		Initial[x] design	Final Design
1	700	A_f	40	102.9
		A_w	360	115.7
		H_w	180	104.1
		A	1100	918.6
2	—	A_f	40	71.1
		A_w	180	29.2
		H_w	80	26.2
		A	260	171.5
3	500	A_f	40	88.1
		A_w	360	157.1
		H_w	180	124.8
		A	900	745.2
4	600	A_f	70	349.3
		A_w	600	330.5
		H_w	250	168.6
		A	1270	1279.8
5	800	A_f	70	311.4
		A_w	700	459.1
		H_w	280	223.1
		A	1570	1570.5
Object function (tons)		$f(x)$	79.58	70.22
No. of analyses				240
No. of unidirectional searches				62
Computer time (seconds)				75.2

[x] Initial design was not feasible, normal stress in member 10 loading condition 1, was 21% above allowable value, and normal stress in member 12, loading condition 1, was 24% above allowable value.

Table 5 Results of Tanker Longitudinal frame Optimization (All dimensions in cm and cm²)

Member type	Plate flage area A_p		Initial design	Final design[x]
1	2000	A_f	300	492.1
		A_w	500	344.4
		H_w	300	139.2
		A	2800	2836.5
2	800	A_f	300	282.3
		A_w	500	650.4
		H_w	300	257.8
		A	1600	1732.7
3	800	A_f	300	225.5
		A_w	500	394.4
		H_w	200	200.7
		A	1600	1419.9
4	Symmetric section	A_f	300	320.7
		A_w	500	523.4
		H_w	300	232.3
		A	1100	1164.8
5	Symmetric section	A_f	300	197.5
		A_w	450	501.3
		H_w	300	236.3
		A	1050	896.3
6	Symmetric section	A_f	200	214.6
		A_w	450	469.3
		H_w	200	226.1
		A	850	898.5
7	1800	A_f	400	186.9
		A_w	600	232.6
		H_w	300	109.1
		A	2800	2219.5
8	700	A_f	100	65.3
		A_w	200	139.2
		H_w	150	117.6
		A	1000	904.5
9	700	A_f	100	55.7
		A_w	200	100.8
		H_w	150	89.1
		A	1000	856.5
Object function (tons)		$f(x)$	115.44	83.30
No. of analyses				361
No. of unidirectional searches				95
Computes time (seconds)				320

[x] Not completely converged

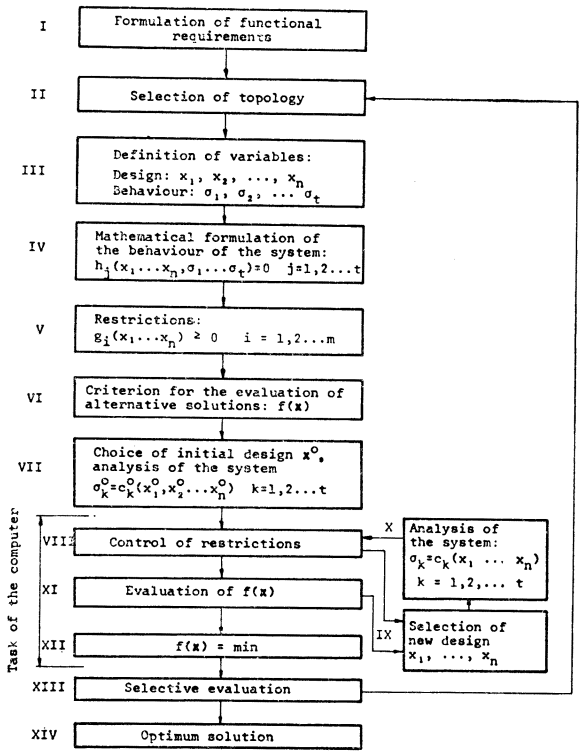


Fig. 1 Schematic outline of the design procedure

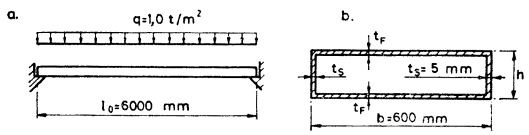
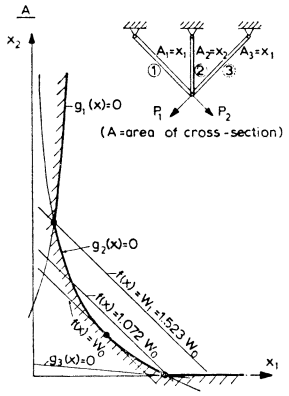


Fig. 2 Hatch cover



Three Bar Truss, Ref. |28|.

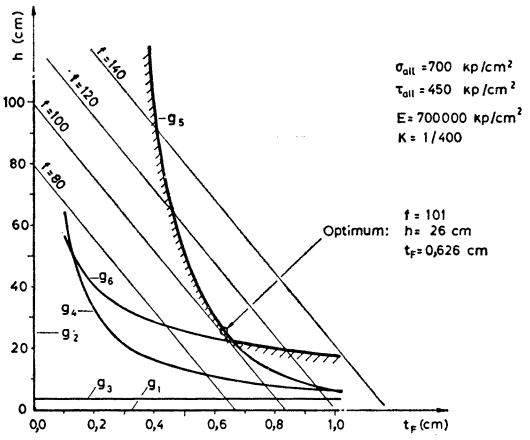
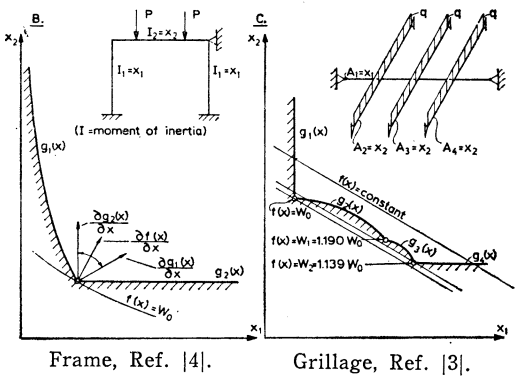


Fig. 3 Restrictions and object function



Frame, Ref. |4|. Grillage, Ref. |3|.

Fig. 4 Simple design examples

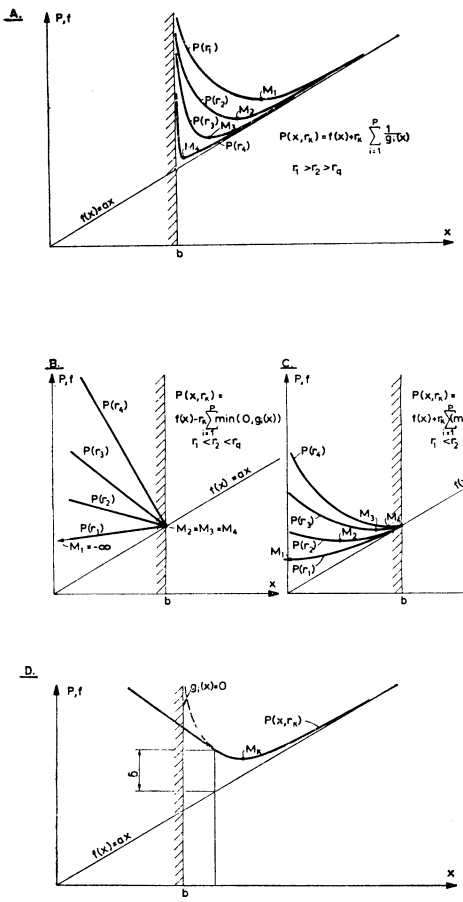


Fig. 5 Some types of penalty functions

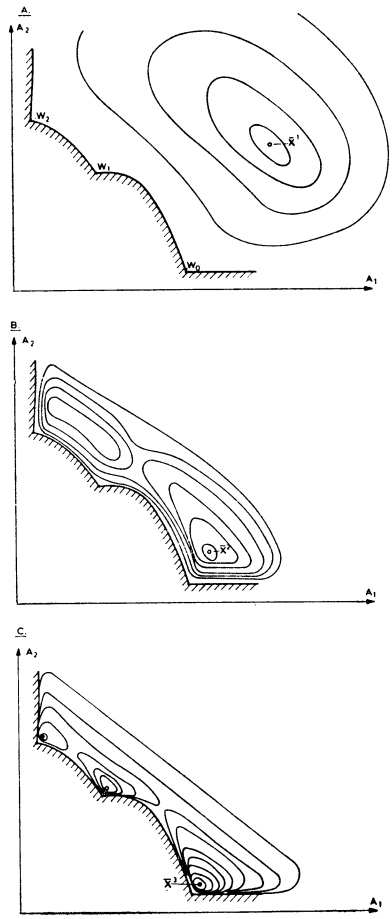


Fig. 6. Appearances of consecutive response surface

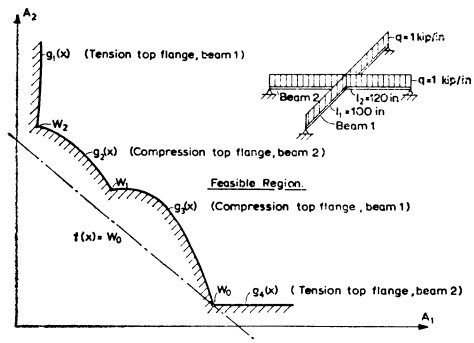


Fig. 7 Simple grillage design problem

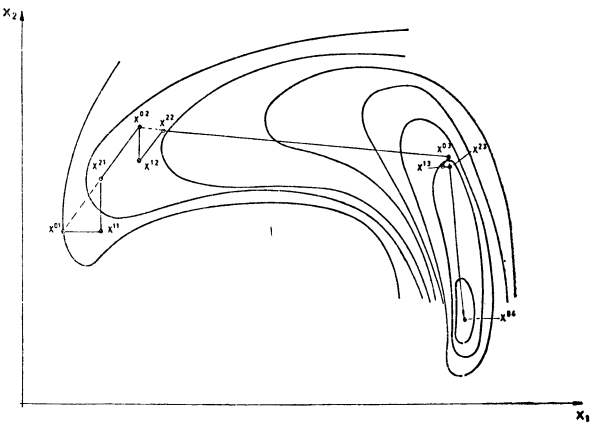


Fig. 8 Powell's method of "direct search"

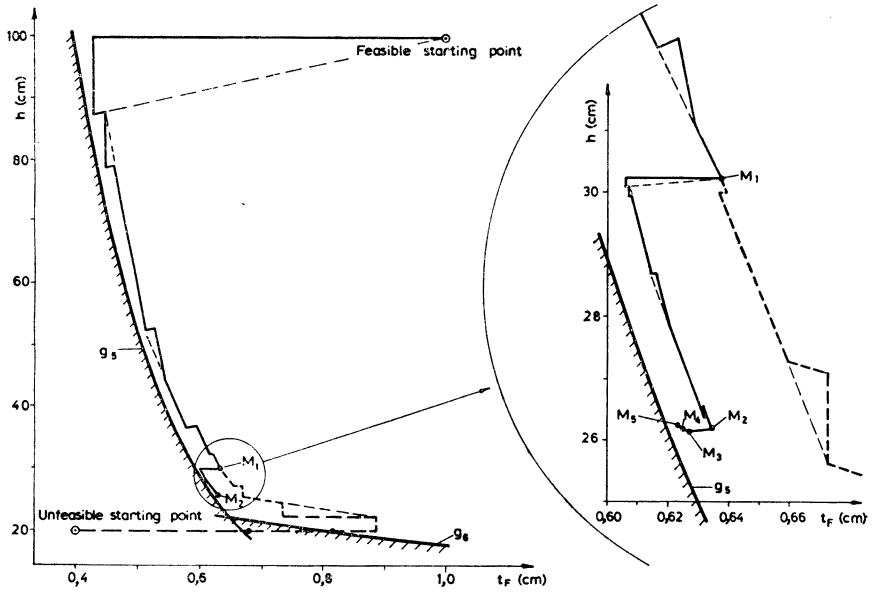


Fig. 9 Patterns of search for example 2.1

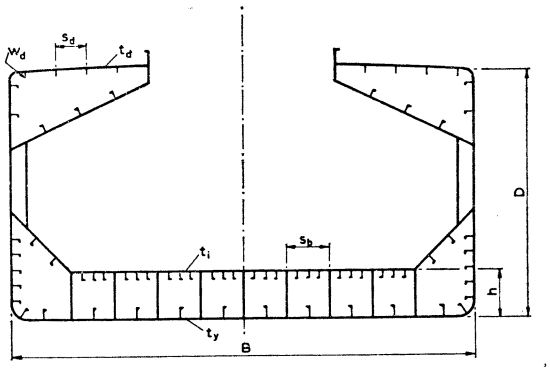


Fig. 10 Midship section of OBO-carrier

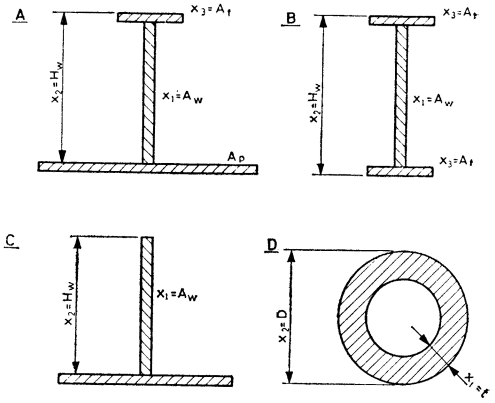


Fig. 11 Typical cross sectional forms

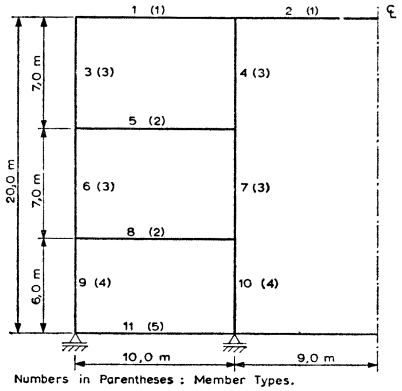


Fig. 12 Transverse web-frame in a tanker

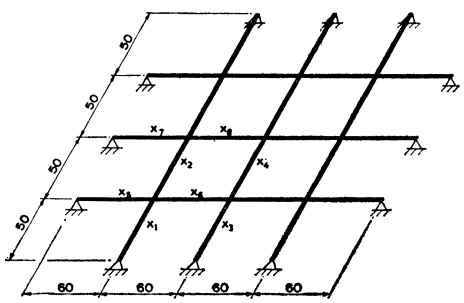


Fig. 13 Grillage design problem

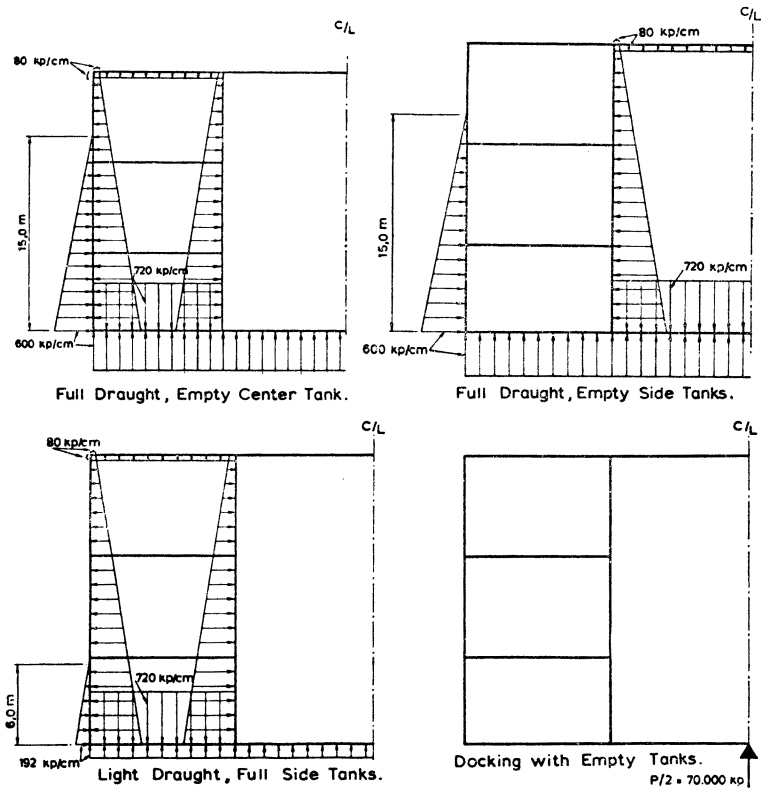


Fig. 14 Loading condition for web-frame

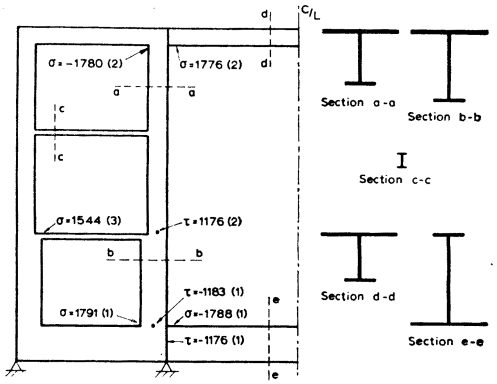


Fig. 15 Results of web frame optimazation

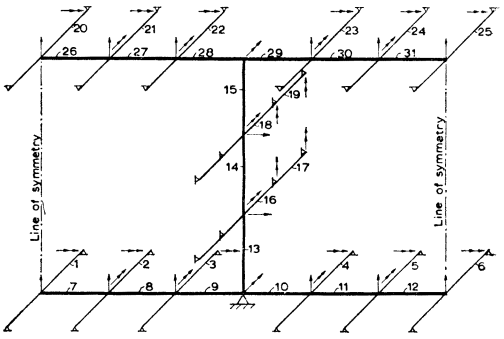


Fig. 16 Longitudinal frame in a tanker